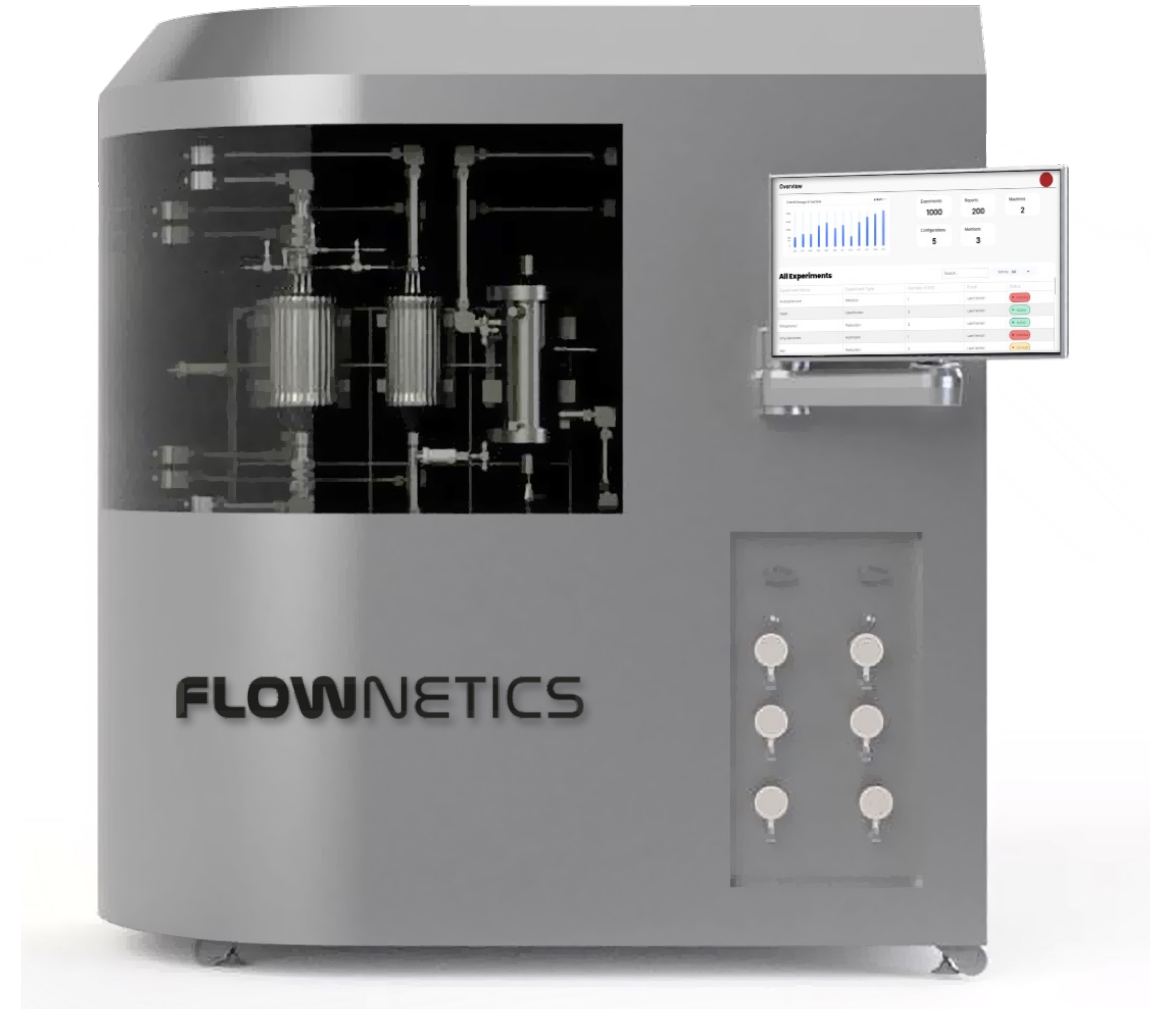
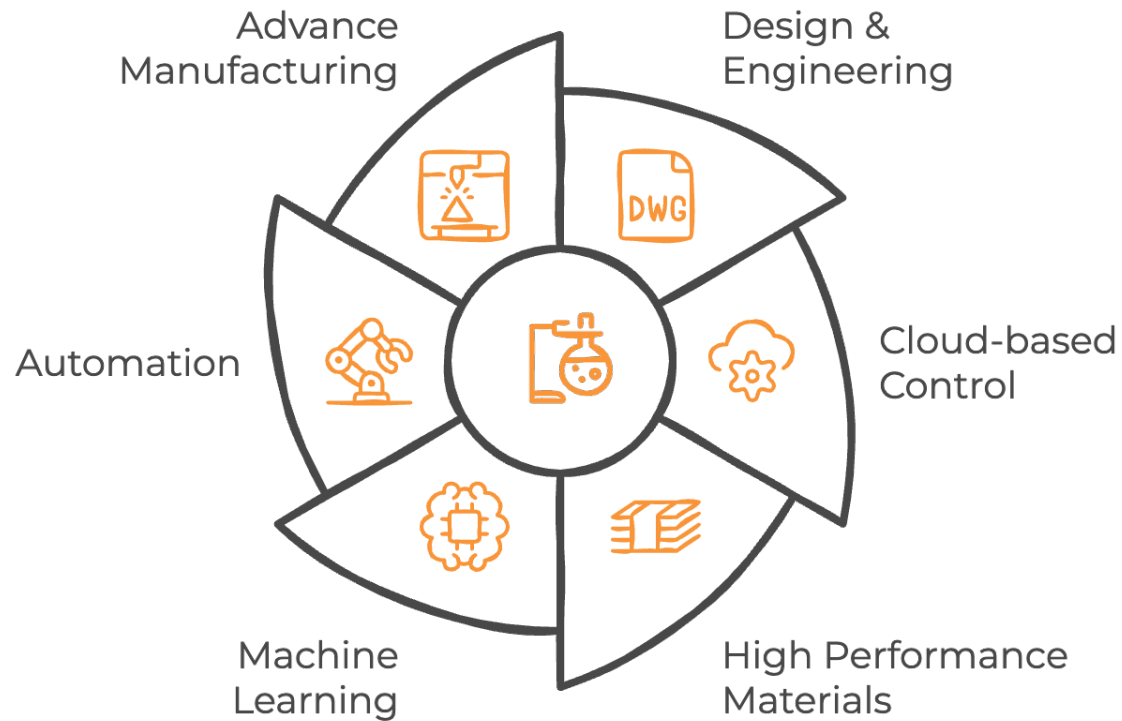


FLOWNETICS

Indigenous Flow Reactor Technology *Capabilities & Use Cases*



*Adding value to your manufacturing processes with our **core competencies** in engineering, product development & continuous flow chemistry*



**Fully-Integrated, Ready-to-Operate, Modular
& Portable Flow Manufacturing System**



REACTOR CHAMBER

- 3 + 1 interchangeable reactors
- 3D Printed in compatible material
- Flexible configurations (telescoping)
- Liquid & gas handling
- Catalyst loading
- Slurry handling
- Up to 100 Bar operating pressure*

Pre-Integrated - All components & utilities required for continuous flow synthesis at your site

FLOWSCALE SOFTWARE SUITE

In-build software for workflow automation and DoE generation

PUMPING STATION

- 4+ Dosing Lines
- In-built liquid dosing pumps
- Gas inlets & MFC
- Slurry dosing

AUTOMATED VALVE ASSEMBLY

Auto/manual switching between solvent and reagent inputs

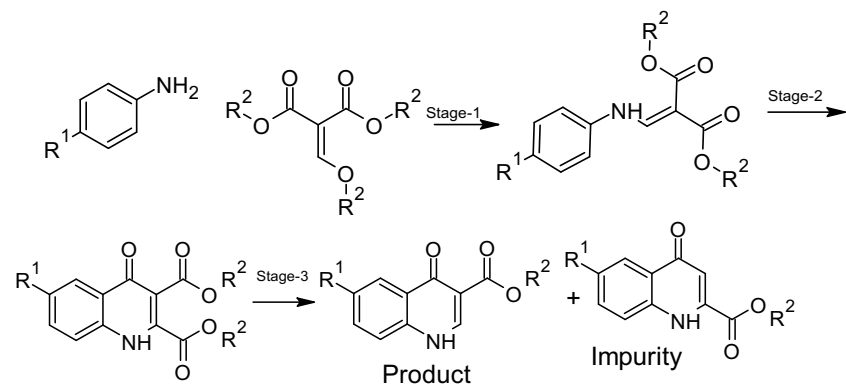


TEMPERATURE CONTROL UNIT

Integrated TCUs with operating range from -90 to 200 °C

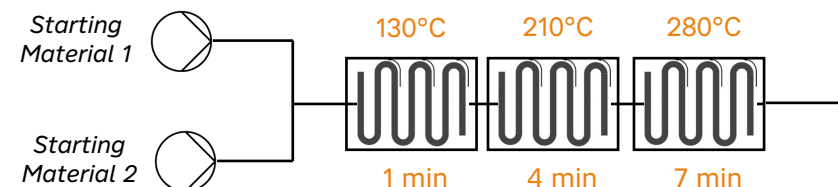


Phase	Description	Process capabilities	Scale-Up Strategy
Phase-1, Tier-1	10 mL, SS316L, 20-25 Bar Liquid-Liquid/ Liquid-Gas Exothermic, Biphasic HPLC Dosing Pumps for Compatible Solutions	This is a liquid -liquid SS platform which is capable of handling solutions, gasses, biphasic liquids that are compatible with SS316L. The reactor options are specifically designed for 1. Homogenous reaction (reactions where starting materials and reagents are in single solvent system or in multiple solvents that are miscible with each other), 2. Biphasic mixing (mass-transfer limited reactions where starting materials and reagents are in multiple solvents or phases like gaseous form that are immiscible with each other) & 3. Handling high exotherms (reactions that require very high temperature control either to maintain a tight window or to remove heat generated during mixing)	1. Lab - 1x 2. Pilot - 10 to 50x (Size Up) 3. Commercial - 200 to 400x (Size Up) 4. Scale-Out (Multiple Modules)
Phase-1, Tier-2	10 mL, SS316L, 20-25 Bar Catalytic Packed-Bed Reactor MFC for Non-Corrosive Gas	This upgrade allows the system to handle heterogenous catalysts in a packed-bed system. It can be used for 1. Liquid-phase reactions with inorganic catalysts & 2. Dose non-corrosive gasses for liquid-gas reactions with or without novel metal catalysts	1. Lab - 1x 2. Pilot - 5 to 25x (Size Up) 3. Commercial - 100 to 200 (Size Up) 4. Scale-Out (Multiple Modules)
Phase-1, Tier-3	10mL, SS316L, 20-25 Bar Solid Handling Peristaltic Dosing Pump for Compatible Slurries	This upgrade allows the system to handle heterogenous reaction where 1. Input stream is liquid and there are chances of solid formation during the reaction & 2. Pump raw materials as slurry (~20%). Note: Depends on the slurry's flow properties and if it doesn't stick or choke the system.	1. Lab - 1x 2. Pilot - 5 to 25x (Size Up) 3. Commercial - 100 to 200 (Size Up) 4. Scale-Out (Multiple Modules)
Phase-2, Tier-1	10 mL, Hastelloy C276/ C22, 20-25 Bar Liquid-Liquid/ Liquid-Gas Exothermic, Biphasic HPLC Dosing Pumps for Compatible Solutions	This is a liquid -liquid SS platform which is capable of handling solutions, gasses, biphasic liquids that are compatible with Hastelloy. The reactor options are specifically designed for 1. Homogenous reaction (reactions where starting materials and reagents are in single solvent system or in multiple solvents that are miscible with each other), 2. Biphasic mixing (mass-transfer limited reactions where starting materials and reagents are in multiple solvents or phases like gaseous form that are immiscible with each other) & 3. Handling high exotherms (reactions that require very high temperature control either to maintain a tight window or to remove heat generated during mixing)	1. Lab - 1x 2. Pilot - 10 to 50x (Size Up) 3. Commercial - 200 to 400x (Size Up) 4. Scale-Out (Multiple Modules)
Phase-2, Tier-2	10 mL, SS316L, Upto 50 Bar & 350 Deg HT/HP Reactor	This upgrade allows the system to handle liquid-liquid reactions upto 350 deg & 50 bar	1. Lab - 1x 2. Pilot - 5 to 25x (Size Up) 3. Commercial - 100 to 200 (Size Up) 4. Scale-Out (Multiple Modules)

Product	Key Starting Material to a low volume API	
Process	Three-step synthesis involving Condensation of secondary amine & diester, followed by Cyclization & Decarboxylation	
Reaction		
Batch Conditions	Stg-1: 120°C for 12hrs Stg-2: 180°C for 16hrs Stg-3: 400°C for 16hrs	80:20 Selectivity 50% Isolated yield
Problem Statement	High temperature, poor selectivity, material loss due to charring & polymerized impurity formation	
Flow Conditions	Telescopic; 130°C, 210°C & 280°C at 12 mins RT	95:5 Selectivity 78% Isolated yield
Objective	One-off requirement of 5 kilos to assist in the drug licence registration, validation & stability studies.	

One-Pot Synthesis

All three reactions telescoped in single platform with multiple high temperature zones



FAST - Development & Execution

28%

Reduction in solvent (high boiling) volume

18%

Increase in Selectivity

56%

Increase in Yield

SECURE

Long-term & localized supply chain

Lab	Campaign (3x)	Alternative
		Three independent batch synthesis 
0.5 g/min	5 Kilos	5 Kilos
~ 3 Weeks	~ 5 days	???

- Three independent batch synthesis

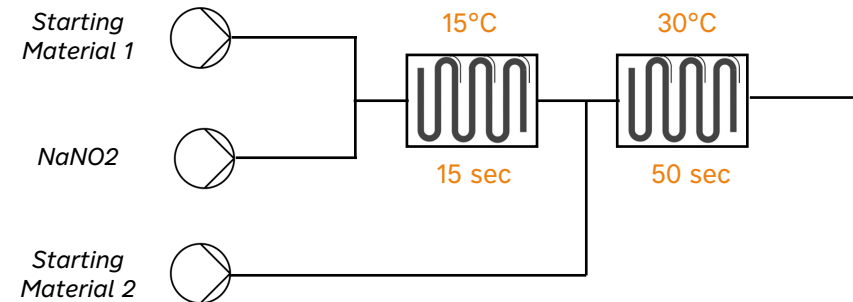


- Multiple batches and repetitive cycles

Product	API Precursor	
Process	Diazotization of an aromatic amine to form highly unstable diazo-intermediate and immediate coupling with second aromatic group	
Reaction	Starting material Product Impurity	
Batch Conditions	Stg-1: - 20°C for 4 hrs Stg-2: 65°C for 3 hrs	76% isolated yield 0.13% single max. impurity (after purification)
Problem Statement	Hazardous, corrosive, exothermic & unstable. Challenging purity profile	
Flow Conditions	Telescopic; : 15°C & 30°C at 1.1 min RT	93 % isolated yield with 0.03% single max. impurity
Objective	Transitioning the current production of 5 tons/month to continuous flow in same plant with better purity profile	

Heat Removed = 16 kW (~ ΔT of +100°C)

No ΔT observed during flow process



SAFE - Handling of chemicals & exotherm

18%
Reduction in Cost

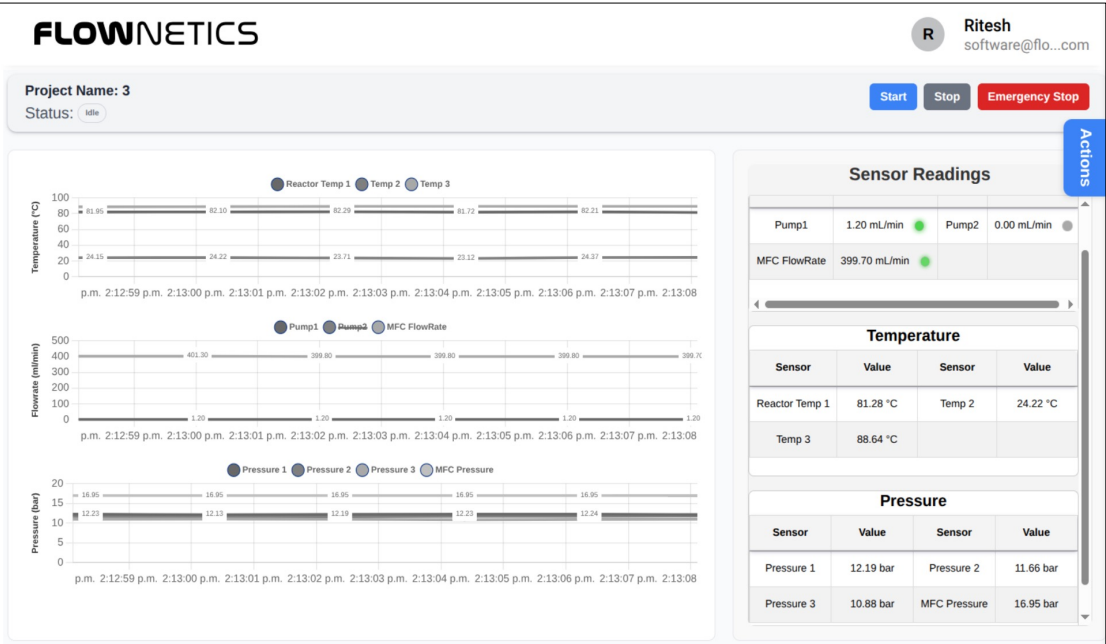
44%
Reduction in Effluents

22%
Increase in Yield

15 months
ROI in Savings

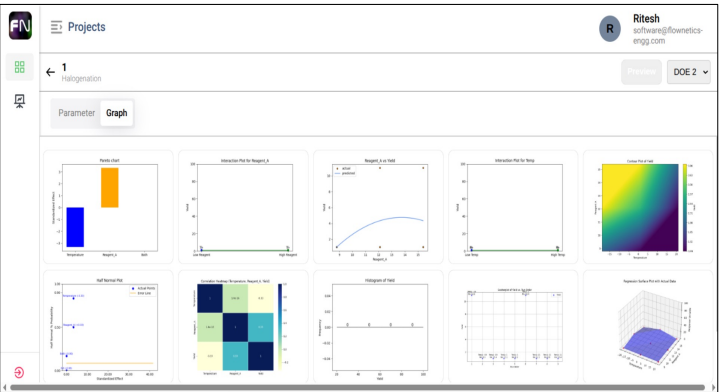
Lab	Pilot (20x)	Plant (100x)
0.1 kg/hr	40 kg/day	5 Tons/Month
~ 3 Weeks	~ 12 Weeks	~ 10 Months

Real-Time
Monitoring &
Machine Control

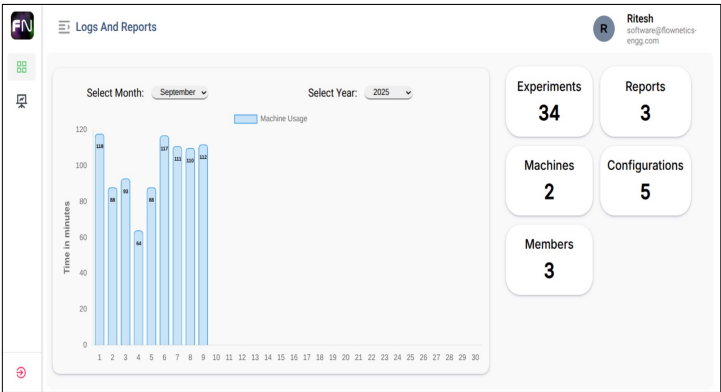


In-House Implementation of 'FlowScale I'

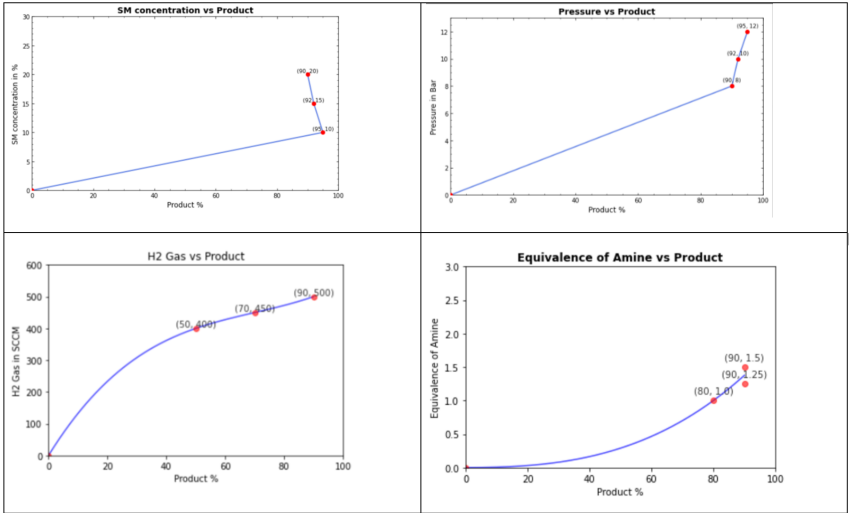
Parameters	Batch conditions	Flow conditions
Temperature	100 C	110 C
Reagent equivalence	1.5	1.25
Pressure	12	8
Residence time	NA	3
Catalyst (% w/w)	0.02	0.004
Concentration	10%	20%
Solvent volume	9	4
Conversion	87-90 %	95%

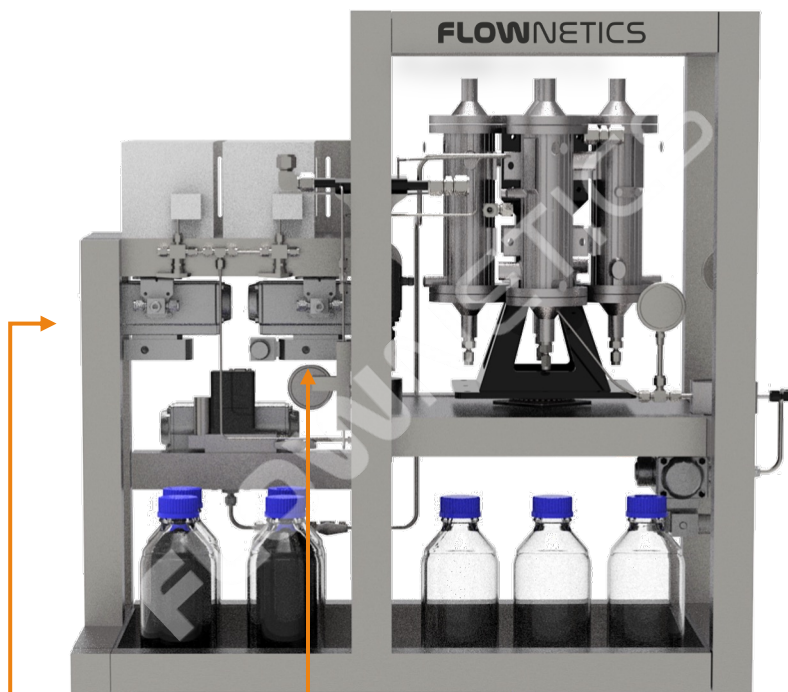


Parameter Optimization &
Data Interpretation



Data Logging &
User Management





DOSING STATION

- 2+ Dosing Lines
- In-built liquid dosing pumps
- Gas inlets & MFC

AUTOMATED VALVE ASSEMBLY

Auto switching between solvent and reagent inputs

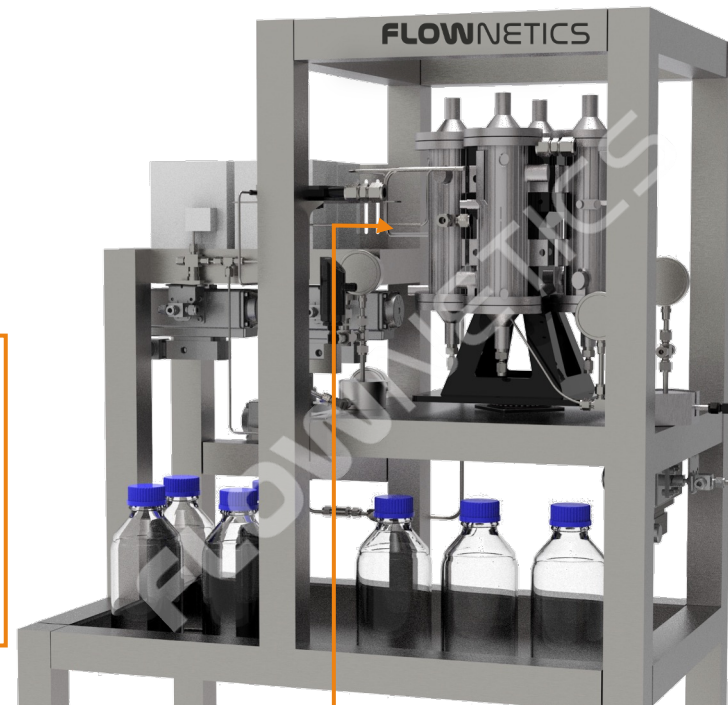


FLOWSCALE SOFTWARE SUITE

In-built software for workflow automation and remote-monitoring

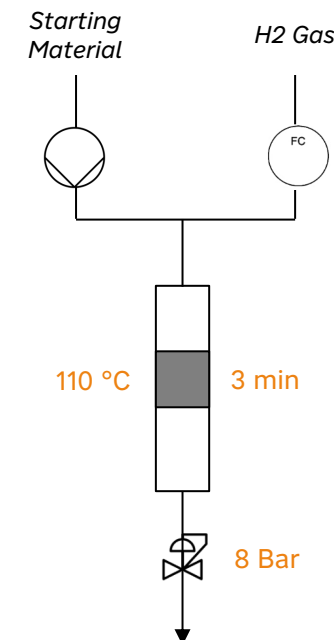
IoT-ENABLED

For Remote & Real-time Monitoring



REACTOR CHAMBER

- 1 + 3 interchangeable reactors
- 3D Printed in compatible metals
- Liquid & Gas handling
- Catalyst loading
- Customizable pressure capability



22%

Reduction in Cost

87%

Reduction in Catalyst Consumption

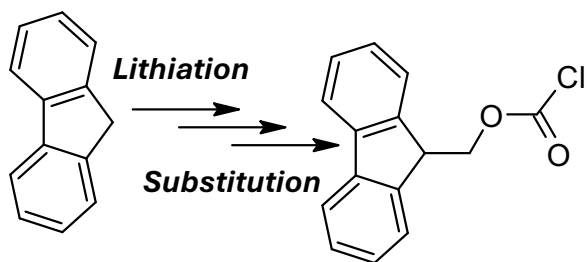
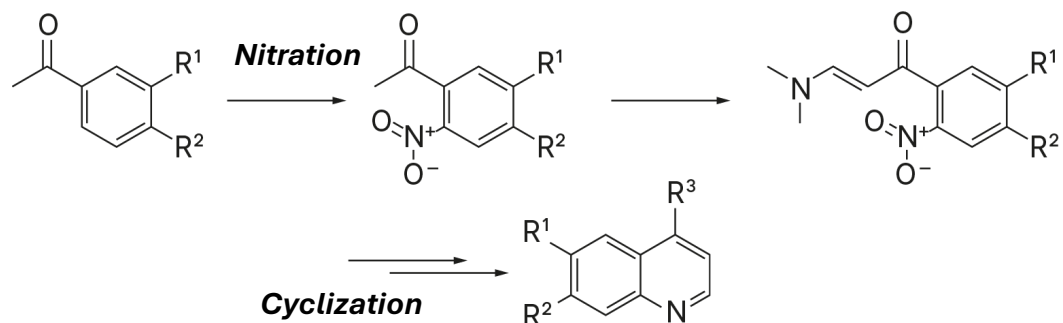
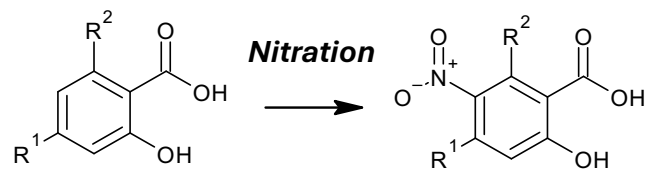
42%

Reduction in Cycle time

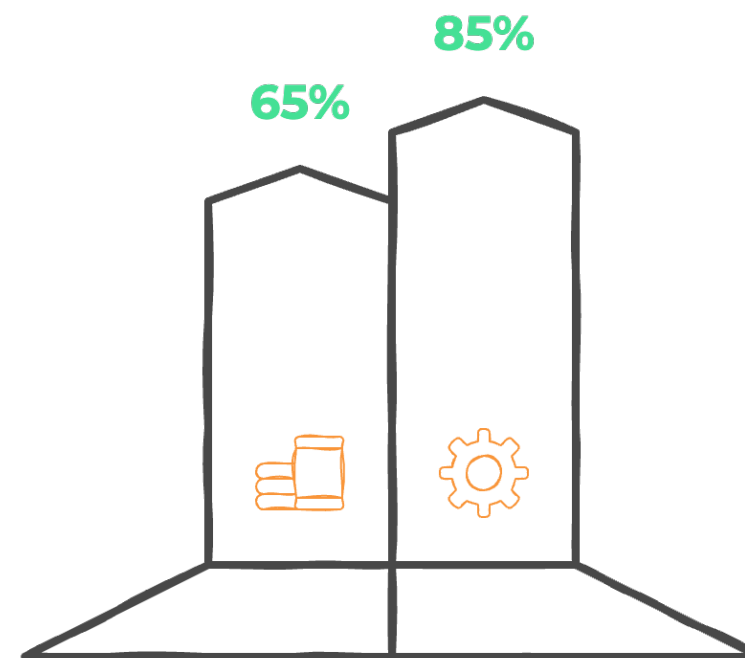
2x

Increase in production capacity same plant

Rapid Screening & Process Optimization : 50+ Experiments in 40 hours!



Savings from Better Yield, Selectivity & Heat Management



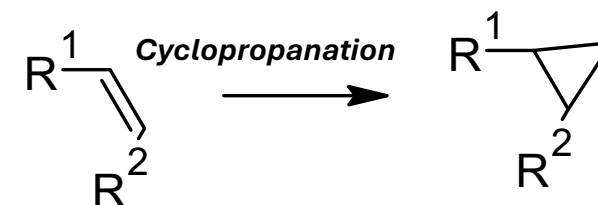
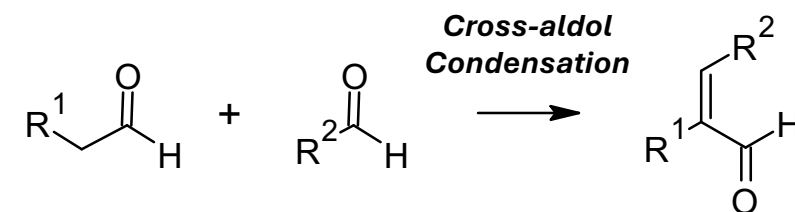
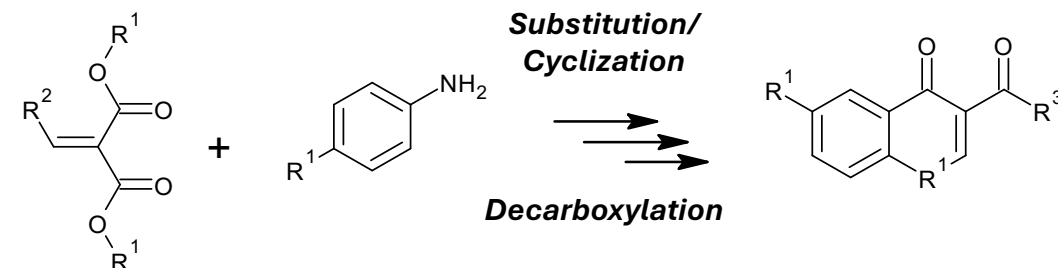
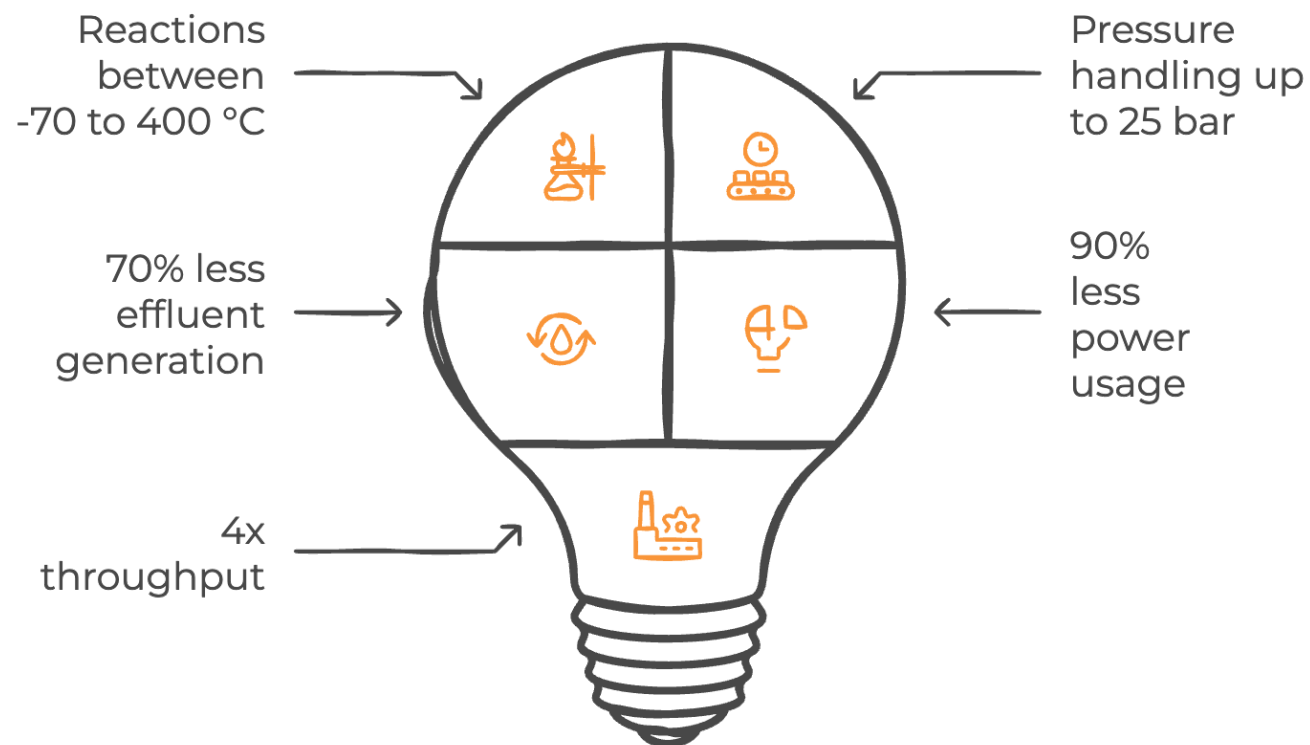
Raw Material Costs

Significant savings on initial inputs

Operational Costs

Major decrease in production expenses

Green Chemistry & Novel Process Windows





Unique
Complex
Problems



Custom
Template
Solutions



One-Stop
~~Multi-Stop~~
Service



OPEX
~~CAPEX~~
Based
Engagement



Continuous
~~One-time~~
Innovation

Where our custom approach has been able to make an impact so far...



EXPANDING CAPACITY
without adding infrastructure



DE-RISKING PRODUCT LAUNCH
with demand-driven expansion



ACCESING NEW MARKETS
with regulatory benefits



LOCALIZING SUPPLY CHAIN
for critical intermediates



REDUCING COMPLEXITY
lesser cycle time & operations



RETAINING COMPETITIVE EDGE
with continuous collab & innovation



SUSTAINABLE MANUFACTURING
by leveraging novel process windows



ELIMINATING HAZARD
with closed handling & automation

Innovative **CAPEX-free** Business Models, Transparent **OPEX-based** Cost Structure

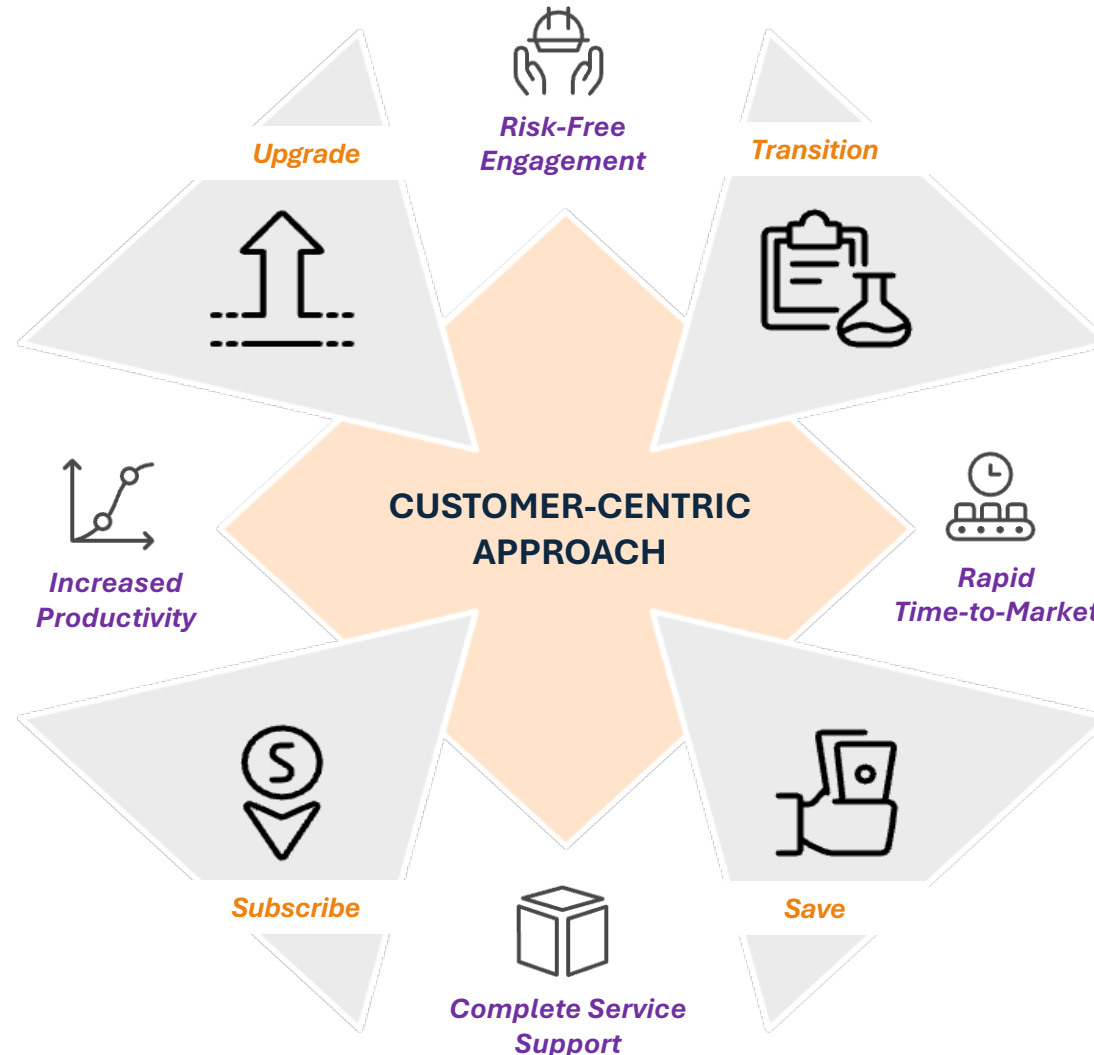
Factory-as-a-Service

Subscription-based continuous manufacturing at customer site



Product-as-a-Service

Exclusive contract manufacturing at Flownetics site

- Development Cost**

One-time service fee for a three-phase R&D program:

 1. Evaluation - Feasibility
 2. Optimization - Intensification
 3. Establishment - Scale Up

Proposed based on number steps, complexity & resources
- Refundable Deposit**

For the equipment to be built and installed at customer site

20-30% of the value of the equipment on a returnable basis
- Subscription Fee**

For operating & maintaining flow equipment on-site

Mutually agreed monthly fee calculated based on new savings & value added
- Next-Gen Advancements**

Exclusive access to software & hardware upgrades developed by Flownetics



First On-Site Customer Engagement
July 2023



Hydrogenation Scale-Up
August 2023



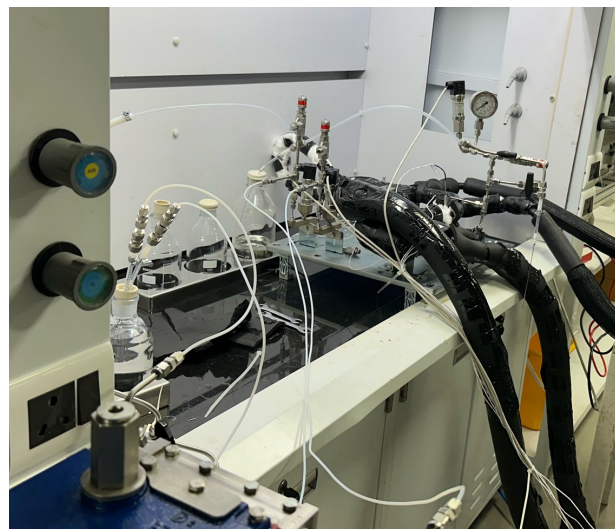
FlowScale Operation Software Suite
September 2023



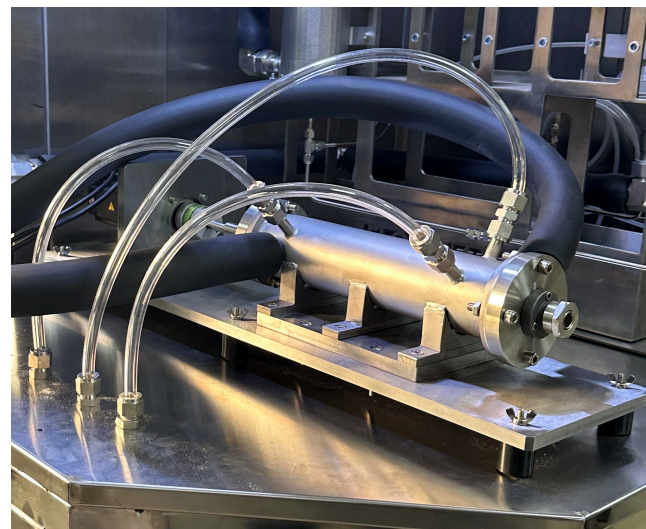
Fully Integrated Hastelloy R&D Platform
June 2024



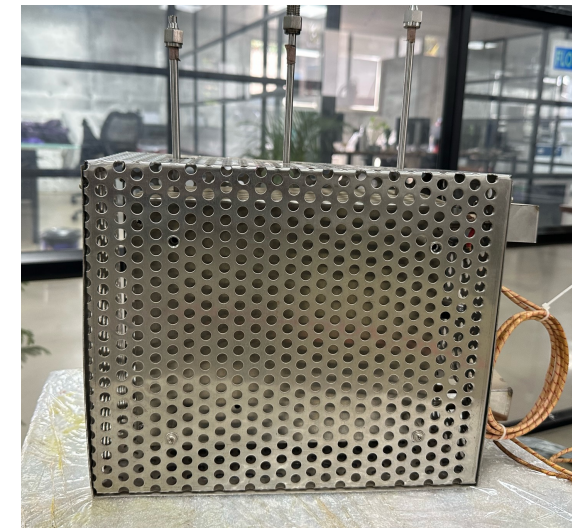
Inauguration of new FlowLab
July 2024



Cryogenic / Pyrophoric Reactions
September 2024



Debating Slurry-phase Flow Reactor
November 2024



High temperature/Pressure Reactions
January 2025

FLOWNETICS

THANK YOU!

